

HouseHunt - Your Perfect Rental Home

1. Introduction

Project Title: HouseHunt - Your Perfect Rental Home

Team ID: LTVIP2026TMIDS65007

Project Memebbers : Vankadara Sai Gayathri, Kumar Kotapati ,Poola Harsha Vardhan
R V S Poojitha

2. Project Overview

HouseHunt is a professional-grade, full-stack web application developed using the MERN ecosystem. It provides a centralized platform for the rental property market, facilitating seamless interaction between property owners and prospective tenants. The application streamlines the process of searching for residential listings through advanced data filtering and secure listing management.

Key Features

- **User Authentication:** Secure registration and login protocols powered by JSON Web Tokens (JWT) and encrypted password storage.
- **Advanced Search & Filtering:** Granular search capabilities including location-based queries, price range (min/max), bedroom count (BHK), property type (Apartment, House), furnished status, and poster category (Owner, Dealer, or Builder).
- **Listing Management:** Comprehensive CRUD (Create, Read, Update, Delete) functionality for authenticated users to manage their rental advertisements.
- **Media Handling:** Integrated property visualization via image URI integration and detailed poster profiles for transparency.

3. Architecture & Tech Stack

The system utilizes a decoupled architecture where the React frontend and Node/Express backend communicate over a RESTful API. This separation of concerns ensures horizontal scalability and modular maintenance.

Detailed Tech Stack

Layer,Technology,Purpose

Frontend,React,Component-based UI library for modular development.

,Axios,Promise-based HTTP client for modular API communication.

,React Hooks,State and lifecycle management within functional components.

Backend,Node.js,Asynchronous event-driven JavaScript runtime.

,Express.js,Minimalist web framework for routing and middleware orchestration.

,Mongoose,Schema-based ODM for MongoDB data modeling.

Database,MongoDB,NoSQL document-oriented storage for flexible data schemas.

Security,JWT,Stateless authentication mechanism for secure claim transmission.

,bcryptjs,Robust one-way hashing for sensitive credential protection.
Environment,dotenv,Management of externalized configuration variables.
,cors,Middleware for governing Cross-Origin Resource Sharing.

4. Setup Instructions

Prerequisites

- **Node.js:** v14.x or later
- **Package Manager:** npm or yarn
- **Database:** MongoDB (Local or Atlas)
- **Version Control:** Git

Backend Installation & Configuration

1. Navigate to the directory: `cd backend`
2. Install dependencies: `npm install`
3. Configure environment: Create a `.env` file in the backend/ root with the following:

```
PORT=5001
MONGO_URI=mongodb://localhost:27017/househuntDB
JWT_SECRET=your_secure_jwt_secret_key_here
```

Frontend Installation & Proxy Details

1. Navigate to the directory: `cd frontend`
2. Install dependencies: `npm install`
3. **Network Configuration:** The React application defaults to port 3000. To resolve Cross-Origin issues during development, the package.json includes a proxy pointing to the backend at `http://localhost:5001`. All relative `/api` calls are automatically routed to the server.

Running the Application

To launch the full stack, execute the following in separate terminal instances:

- **Server Side:** `cd backend && npm run server`
- **Client Side:** `cd frontend && npm start`

Data Seeding

For testing purposes, use the built-in seeding utility:

- **Populate DB:** `npm run seed` (Clears existing records and inserts sample users/listings).
- **Wipe DB:** `npm run seed:destroy` (Removes all records from users and listings collections).

5. Folder Structure

Backend Organization

- `config/`: Logic for database connectivity (`db.js`).

- controllers/: Request handling logic; interacts with models to process data.
- middleware/: Request Interceptors, including authMiddleware.js for JWT validation.
- models/: Mongoose schemas defining the data structure for Users and Listings.
- routes/: Definition of API endpoints and mapping to specific controllers.
- utils/: Helper scripts, including the database seeder tool.

Frontend Organization

- src/services/: Contains modularized Axios instances and API abstraction logic.
- src/components/Auth/: Manages user lifecycle views (Login, Registration).
- src/components/Filters/: State-driven components for refining listing search results.
- src/components/Layout/: Global structural components like Navigation and Footers.
- src/components/Listings/: Core logic for rendering, creating, and editing property data.

6. API Documentation

User Authentication

POST /api/users/register

- **Description:** Registers a new account.
- **Request Body:** { "name": "John Doe", "email": "john@example.com", "password": "password123" }
- **Success Response:** 201 Created | { "token": "JWT_STRING", "user": { "id": "...", "name": "..." } }

POST /api/users/login

- **Description:** Authenticates user and grants access.
- **Request Body:** { "email": "john@example.com", "password": "password123" }
- **Success Response:** 200 OK | { "token": "JWT_STRING", "user": { "id": "...", "name": "..." } }

Listing Management

GET /api/listings

- **Description:** Fetches rental listings with optional query parameters.
- **Parameters:** location, bhk, maxPrice, propertyType.
- **Success Response:** 200 OK | { "title": "Modern Flat", "price": 1200, ... }

POST /api/listings

- **Description:** Creates a new rental ad. (Requires x-auth-token header).
- **Request Body:** { "title": "House for Rent", "location": "City Center", "price": 2000, "bhk": 3, "image": "URL" }
- **Success Response:** 201 Created | { "msg": "Listing created successfully" }

PUT /api/listings/:id

- **Description:** Updates a listing. (Requires x-auth-token and ownership).
- **Request Body:** { "price": 1800 }
- **Success Response:** 200 OK | { "updatedListing": { ... } }

DELETE /api/listings/:id

- **Description:** Removes a listing. (Requires x-auth-token and ownership).
- **Success Response:** 200 OK | { "msg": "Listing removed" }

7. Authentication

Security Logic

The application utilizes a stateless authentication strategy via **JSON Web Tokens (JWT)** . Sensitive credentials undergo **one-way hashing** using the bcryptjs algorithm before being persisted to the database, ensuring that plain-text passwords are never stored.

Authentication Flow

1. **Handshake:** During login, the server compares the provided password with the stored hash.
2. **Issuance:** Upon verification, the backend issues a signed JWT containing the user's ID.
3. **Client Storage:** The frontend receives the token and stores it in localStorage for session persistence.
4. **Authorization:** For protected routes, the client includes the token in the x-auth-token header. The authMiddleware.js on the server intercepts these requests, verifies the token's integrity, and permits or denies access based on the validation result.

8. Screenshots or Demo

Interface Overview

HouseHunt

LoginRegister

Find Your Home

Location:

e.g., Mumbai, Delhi

Min Price (₹/month):

0

Max Price (₹/month):

Any

Bedrooms (BHK):

Any

Property Type:

Any

Furnished Status:

Any

Listed By:

Available Rentals



Royal house

Location: tirupati

₹15,000/month

3 BHK |2 Bath(s) |10000 sq ft

Type: Apartment |Furnished: Unfurnished

Posted by: Royal (Listed as: Owner)



Pavan house

Location: Tirupati

₹15,000/month

2 BHK |2 Bath(s) |1500 sq ft

Type: Apartment |Furnished: Unfurnished

Posted by: Pavan (Listed as: Owner)



Vamsi House

Location: Kodur

₹5,00,000/month

4 BHK |1 Bath(s) |183539 sq ft

Type: House |Furnished: Unfurnished

Posted by: vamsi (Listed as: Owner)

HouseHunt

LoginRegister

Find Your Home

Location:

e.g., Mumbai, Delhi

Min Price (₹/month):

0

Max Price (₹/month):

Any

Bedrooms (BHK):

Any

Property Type:

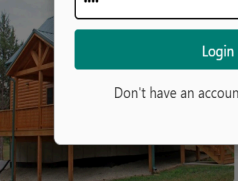
Any

Furnished Status:

Any

Listed By:

Available Rentals



Royal house

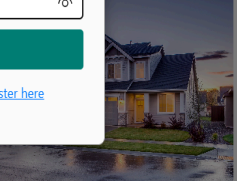
Location: tirupati

₹15,000/month

3 BHK |2 Bath(s) |10000 sq ft

Type: Apartment |Furnished: Unfurnished

Posted by: Royal (Listed as: Owner)



Pavan house

Location: Tirupati

₹15,000/month

2 BHK |2 Bath(s) |1500 sq ft

Type: Apartment |Furnished: Unfurnished

Posted by: Pavan (Listed as: Owner)



Vamsi House

Location: Kodur

₹5,00,000/month

4 BHK |1 Bath(s) |183539 sq ft

Type: House |Furnished: Unfurnished

Posted by: vamsi (Listed as: Owner)

Login

Username:

Vamsi

Password:

....

Login

Don't have an account? [Register here](#)

Furnished Status:

Any

Listed By:

Any

Search Homes

Reset Filters

Post New Ad

Type: Apartment |Furnished: Unfurnished

Posted by: Royal (Listed as: Owner)

this is Royal house

Type: Apartment |Furnished: Unfurnished

Posted by: Pavan (Listed as: Owner)

this is pavan house

Type: House |Furnished: Unfurnished

Posted by: vamsi (Listed as: Owner)

this is my house, please call me



Spacious 3 BHK in Koramangala

Location: Bangalore

₹65,000/month

3 BHK |3 Bath(s) |1800 sq ft

Type: Apartment |Furnished: Semi-Furnished

Posted by: rajeshK (Listed as: Owner)

Well-ventilated 3 BHK apartment with



Cozy 1 BHK Studio, Andheri West

Location: Mumbai

₹32,000/month

1 BHK |1 Bath(s) |650 sq ft

Type: Apartment |Furnished: Furnished

Posted by: priyaS (Listed as: Owner)

Fully furnished studio apartment perfect for



Independent House, Jubilee Hills

Location: Hyderabad

₹90,000/month

4 BHK |4 Bath(s) |3200 sq ft

Type: House |Furnished: Unfurnished

Posted by: amitV (Listed as: Owner)

Large independent house with a beautiful

Well-ventilated 3 BHK apartment with modern amenities, car parking, and close to tech parks. Ideal f...



Modern 2 BHK, Powai Lake View

Location: Mumbai

₹72,000/month

2 BHK |2 Bath(s) |1150 sq ft

Type: Apartment |Furnished: Furnished

Posted by: rajeshK (Listed as: Dealer)

Luxurious 2 BHK in a premium complex with lake view. Includes modular kitchen and society amenities.

Fully furnished studio apartment perfect for singles. Walking distance to Andheri station. Includes ...



Builder Floor 3BHK, GK-1

Location: Delhi

₹58,000/month

3 BHK |2 Bath(s) |1600 sq ft

Type: Condo |Furnished: Semi-Furnished

Posted by: sunitaM (Listed as: Builder)

Newly constructed builder floor with lift and stilt parking. Spacious rooms and good quality fitting...

Large independent house with a beautiful private garden and ample parking. Prime, peaceful locality.

Find Your Home

Location:

Tirupati

Min Price (₹/month):

15000

Max Price (₹/month):

30000

Bedrooms (BHK):

Any

Property Type:

Any

Furnished Status:

Any

Listed By:

Available Rentals



Royal house

Location: tirupati

₹15,000/month

3 BHK | 2 Bath(s) | 10000 sq ft

Type: Apartment | **Furnished:** Unfurnished

Posted by: Royal (Listed as: Owner)



Pavan house

Location: Tirupati

₹15,000/month

2 BHK | 2 Bath(s) | 1500 sq ft

Type: Apartment | **Furnished:** Unfurnished

Posted by: Pavan (Listed as: Owner)

Register

Username:

Royal

Password:

Confirm Password:

Register

Already have an account? [Login here](#)

Find Your Home

Location:

Tirupati

Min Price (₹/month):

15000

Max Price (₹/month):

30000

Bedrooms (BHK):

Any

Property Type:

Any

Furnished Status:

Any

Listed By:

Available Rentals



Royal house

Location: tirupati

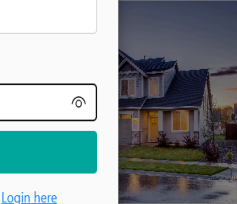
₹15,000/month

3 BHK | 2 Bath(s) | 10000 sq ft

Type: Apartment | **Furnished:** Unfurnished

Posted by: Royal (Listed as: Owner)

this is Royal house



Pavan house

Location: Tirupati

₹15,000/month

2 BHK | 2 Bath(s) | 1500 sq ft

Type: Apartment | **Furnished:** Unfurnished

Posted by: Pavan (Listed as: Owner)

this is pavan house



Vamsi House

Location: Kodur

₹5,00,000/month

4 BHK | 1 Bath(s) | 183539 sq ft

Type: House | **Furnished:** Unfurnished

Posted by: vamsi (Listed as: Owner)

this is my house, please call me

9. Future Enhancements

- **Real-time Messaging:** Implementation of a WebSockets-based chat system to facilitate direct negotiation between tenants and landlords.
- **Cloud-based Asset Management:** Migration from URL-based images to a dedicated storage solution like **Cloudinary** or **AWS S3** for direct file uploads.
- **Geospatial Integration:** Integration of the Google Maps API for interactive map-based search and property proximity visualization.